5580 Data and Text Mining

Master of Science in Computing and Data Analytics

Department of Mathematics and Computing Science

NSPI Electricity Analytics: Forecasting and Anomaly Detection

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Abstract / Executive Summary

Nova Scotia Power (NSPI) is operating in a period of demand volatility driven by electrification, weather variability, and shifting load behavior. This report presents an integrated workflow combining regional forecasting and consensus-based anomaly detection to improve planning confidence and operational triage.

1 Data Exploration & Engineering

We analyze 438,240 hourly records (2015–2024) across Annapolis Valley, Cape Breton, Halifax, Pictou County, and South Shore. Inputs combine demand, weather, and calendar context.

1.1 Dataset profile

The dataset covers roughly 2015–2024 (hourly) for Annapolis Valley, Cape Breton, Halifax, Pictou County, and South Shore. Demand, weather, and calendar context are integrated; all models are evaluated per region.

Regional heterogeneity is substantial. Halifax carries the largest load amplitude and stronger peak patterns, while other regions present lower baselines but sharper relative excursions during weather events. This makes pooled global baselines less reliable for operations and justifies region-wise model fitting and evaluation.

1.2 Data quality handling

Preparation focused on reliability:

- dropped invalid timestamps and missing region IDs,
- standardized numeric fields,
- filled missing values with region-aware interpolation/means,
- retained outliers for anomaly analysis.

This preprocessing strategy preserves operational signal instead of over-smoothing the data. In power-system monitoring, rare spikes are often the events of interest; removing them during cleaning can hide reliability-relevant behavior and reduce downstream detection sensitivity.

1.3 Feature engineering

Features capture autocorrelation, trend, volatility, weather, and seasonality:

- **Lag features:** `consumption_kwh_lag_1h`, `consumption_kwh_lag_24h`
- **Rolling statistics:** `rolling_mean_168h`, `rolling_std_24h`
- **Weather/time bins:** temperature, humidity, hour, day-of-week, weekend/holiday indicators

Detailed diagnostics are in the appendix (Figure 7, Figure 8, Figure 9).

From a forecasting perspective, lag and rolling terms stabilize short-horizon predictions by providing persistence and local-trend context. From an anomaly perspective, volatility

and weather-linked variables improve contextual interpretation (for example, separating demand spikes caused by weather stress from potentially abnormal load behavior).

2 Forecasting Methodology (Task 2A)

2.1 Modeling strategy

Two forecasting approaches were evaluated for 7-day operational planning support:

- **Prophet**: additive/multiplicative seasonality with exogenous regressors.
- **Advanced-LSTM-Optuna**: multivariate sequence learning with lag, rolling, weather, and calendar features, tuned with Optuna.

Hyperparameter tuning used an Optuna-style search process to improve generalization and reduce regional under/over-fitting. Tuning targets included sequence configuration, hidden dimension, learning rate, and regularization-related settings.

For each region, data was sorted chronologically and split into train/test windows to preserve temporal causality. This avoids leakage from future observations and better reflects real operational deployment, where future load values are unavailable at prediction time. All comparisons were performed on aligned test horizons to keep MAE, RMSE, and MAPE directly comparable.

2.2 Forecast metrics comparison

Forecast quality was compared using MAE, RMSE, and MAPE by region.

Expanded metric details are also provided in the appendix (see Table 1), while extended visual diagnostics are included in Figure 5 and Figure 6.

A direct forecast comparison plot (actual vs Prophet vs Advanced-LSTM-Optuna) is provided in Figure 1.

Metric interpretation follows complementary roles: MAE captures average absolute miss, RMSE emphasizes large misses, and MAPE provides scale-normalized interpretability for cross-region discussion. Using all three prevents over-optimizing one behavior (e.g., average error) at the expense of tail-risk behavior that matters for peak operations.

2.3 Business interpretation

The neural model outperformed Prophet in four regions, while Prophet remained stronger in Halifax. This indicates that NSPI should adopt region-specific model champions, rather than enforcing a single forecasting model across all geographies.

Related feature-space diagnostics that inform model behavior are provided in Figure 7, Figure 8, and Figure 9.

Operationally, this supports a champion-challenger governance model: maintain one selected production model per region, but continuously benchmark alternatives as demand behavior evolves seasonally. This approach balances reliability, maintainability, and model performance over time.

3 Anomaly Detection (Task 2B)

3.1 4-Tier ensemble design

Anomaly detection uses a layered ensemble to capture complementary abnormality types:

- **Z-Score**: global spikes,
- **IQR**: robust outliers,
- **Isolation Forest**: contextual anomalies,
- **LSTM Autoencoder**: temporal anomalies.

The design intentionally combines statistical, distributional, contextual, and temporal detectors. Single-method systems often produce unstable alert streams under changing weather and load regimes; ensemble diversity helps maintain alert quality when one detector becomes locally less reliable.

3.2 Mathematical definitions

Z-score:

$$z = \frac{x - \mu}{\sigma}$$

Autoencoder reconstruction loss:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{x}_i)^2$$

3.3 Calibration and consensus

Method outputs were harmonized with a shared target anomaly rate for Isolation Forest contamination and autoencoder fallback quantiles. Autoencoder thresholds use median-plus-MAD first, then quantile fallback when rates move outside the operational band.

This calibration strategy controls alert-volume drift across regions and periods. In practice, operations teams need predictable alert volumes to avoid triage overload; bounded calibration keeps the system actionable while preserving sensitivity to meaningful deviations.

Final escalation uses a high-confidence consensus rule:

- **Anomaly flag = True** when two or more methods agree.

This reduces false positives and improves triage.

3.4 Validation

Detected anomalies were compared against `anomaly_flag`. Consensus alerts improved precision over single-method triggers while keeping useful recall.

Validation was treated as operational rather than purely academic: the objective is to surface incidents worth investigation, not to maximize raw anomaly counts. Consensus rules therefore prioritize higher-confidence incidents for dispatch and root-cause analysis.

For supporting visuals, see the appendix: final 4-method comparison in Figure 2, training diagnostics in Figure 3, and reconstruction diagnostics in Figure 4. Aggregated method totals are summarized in Table 2 and Table 3.

4 Tableau Insights

4.1 Dashboard layout strategy

The dashboard is structured for fast escalation from overview to incident details:

- KPI summary (total, consensus, anomaly rate),
- region-method heatmap and trend slices,
- method comparison across the four detectors,
- row-level drill-down with method votes and consensus.

The layout supports two user personas. Executives use KPI and trend panels to assess system-level risk posture, while analysts and operators use drill-down views to inspect specific timestamps, regions, and detector agreement patterns.

4.2 Business value for NSPI

This layout supports faster reliability decisions by helping teams:

- prioritize high-volatility regions,
- track peak/weather stress concentration,
- separate weak alerts from high-confidence consensus events.

In practice, this improves prioritization discipline. Incidents with multi-method agreement and high local concentration can be elevated quickly, while isolated single-detector events can be monitored before costly intervention.

4.3 Explainable AI tooltips

Tooltips expose triggering methods, vote counts, and anomaly context so operators can quickly understand **why** a point was flagged.

4.4 Published data products

The Tableau pipeline uses:

- `anomaly_results.csv`
- `autoencoder_results.csv`
- `region_method_summary.csv`
- `global_method_summary.csv`

These outputs support both incident triage and executive monitoring.

The split between row-level and aggregate products is intentional: row-level outputs support forensic analysis and reproducibility, while aggregate products provide concise trend communication for recurring operational reviews.

Additional drill-down visuals are in the appendix: Figure 10, Figure 11, Figure 12, and

Figure 13.

These outputs support both incident triage and executive monitoring.

5 Conclusion & Recommendations

This project delivers an end-to-end NSPI workflow from engineered features to forecasting and explainable anomaly detection. Forecast outputs support planning, while consensus anomaly alerts support operational triage.

5.1 Key conclusions

- Regional demand behavior differs enough to require region-specific models.
- Neural forecasting leads in most regions; Prophet remains competitive in Halifax.
- The 4-method ensemble captures complementary anomaly patterns.
- Consensus alerts (2+ methods) reduce false positives and improve actionability.

5.2 Recommendations for NSPI

1. Use region-specific forecast champions selected by local MAE/RMSE/MAPE.
2. Use consensus anomaly flags as the default operational alert threshold.
3. Prioritize incidents where contextual and temporal detectors agree.
4. Recalibrate thresholds seasonally to keep alert volumes stable.
5. Continue validating model outputs against labels and field outcomes.

Implementation can follow a phased rollout: first deploy dashboarding with existing outputs, then operationalize consensus thresholds for alerting, and finally establish periodic retraining/calibration windows. This creates a manageable adoption path with measurable reliability improvements at each stage.

5.3 Operational rollout roadmap

A practical rollout can be organized into three stages:

- **Stage 1 – Visibility**: publish Tableau dashboards using the current exported datasets and establish weekly monitoring reviews.
- **Stage 2 – Controlled alerting**: enable consensus-based anomaly escalation for analyst triage with clear escalation playbooks.
- **Stage 3 – Continuous improvement**: schedule retraining, threshold recalibration, and periodic backtesting against new operational outcomes.

This phased approach reduces change risk while providing immediate value. It also supports auditability because each stage has a clear data source, evaluation checkpoint, and ownership model.

5.4 Limitations and future work

The current system is strong for short-horizon planning and operational anomaly triage, but several enhancements can improve production readiness: richer external regressors (storm alerts and outage feeds), explicit probabilistic forecasting intervals, and event-level

postmortem labels to strengthen validation quality.

Future work should also evaluate lead-time sensitivity (how early anomalies are detected before major events) and intervention utility (how many alerts produce actionable preventive decisions). These two dimensions are critical for measuring business value beyond traditional model-fit metrics.

These actions provide a practical path to preserve reliability while supporting NSPI's clean-energy transition.

6 Appendix

6.1 Project Repository

GitHub repository

6.2 Supplementary Tables

Table 1: Forecasting metrics by region and model (latest notebook run)

Region	Prophet			Advanced-LSTM-Optuna		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
Annapolis Valley	20.16	48.99	22.00	25.43	52.22	26.00
Cape Breton	36.36	85.21	20.00	45.70	91.09	21.00
Halifax	93.99	213.15	23.00	131.58	238.77	25.00
Pictou County	12.56	28.13	28.00	18.69	32.20	37.00
South Shore	17.07	39.34	27.00	20.83	41.97	33.00

Table 2: Global anomaly detection summary by method

Method	Total Anomaly	% of Total Rows	% within Method Flags
IQR	6596	1.51	19.60
Isolation Forest	8765	2.00	26.05
LSTM Autoencoder	14912	3.40	44.32
Z-Score	3373	0.77	10.02

Table 3: Region-wise anomaly summary (total anomalies by method)

Region	Z-Score	IQR	Isolation Forest	LSTM Autoencoder
Annapolis Valley	720	1093	1753	3151
Cape Breton	692	1644	1753	2698
Halifax	687	1304	1753	2981
Pictou County	601	1378	1753	2934
South Shore	673	1177	1753	3148

6.3 Supplementary Figures from Modeling Outputs

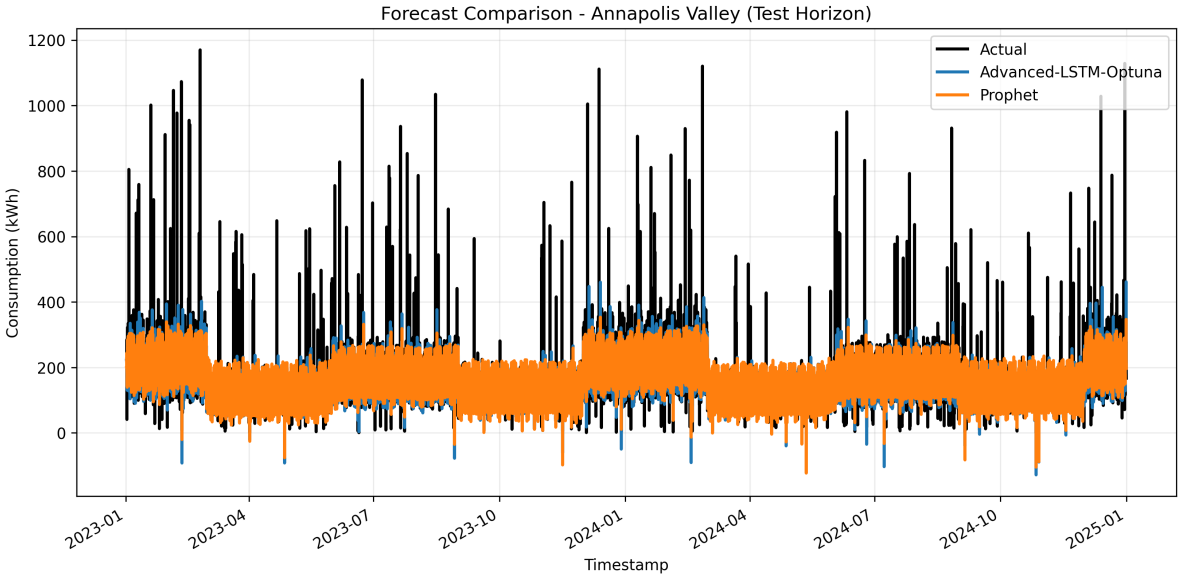

Figure 1: Forecast comparison of actual consumption vs Prophet and Advanced-LSTM-Optuna (test horizon).

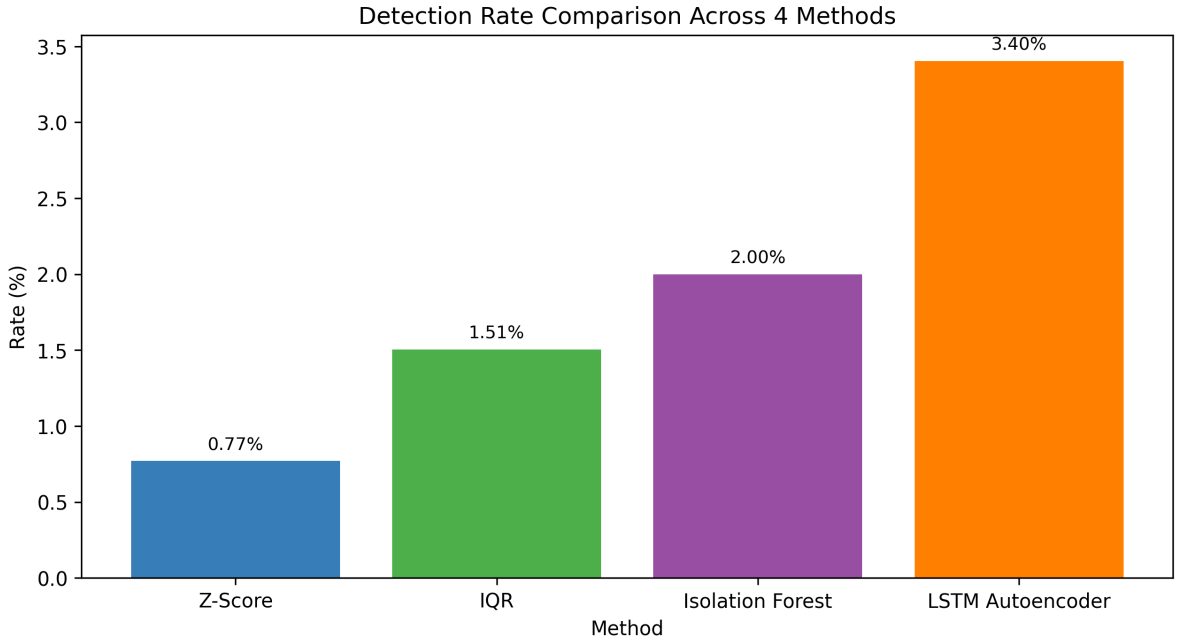

Figure 2: Final 4-method anomaly detection rate comparison.

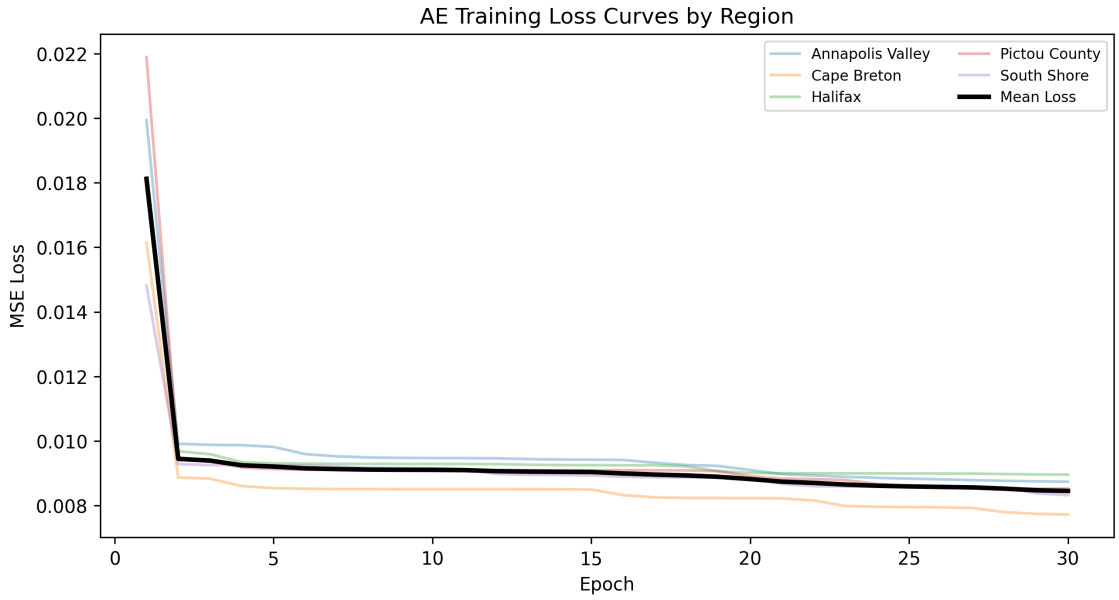

Figure 3: Autoencoder training loss curves across regions.

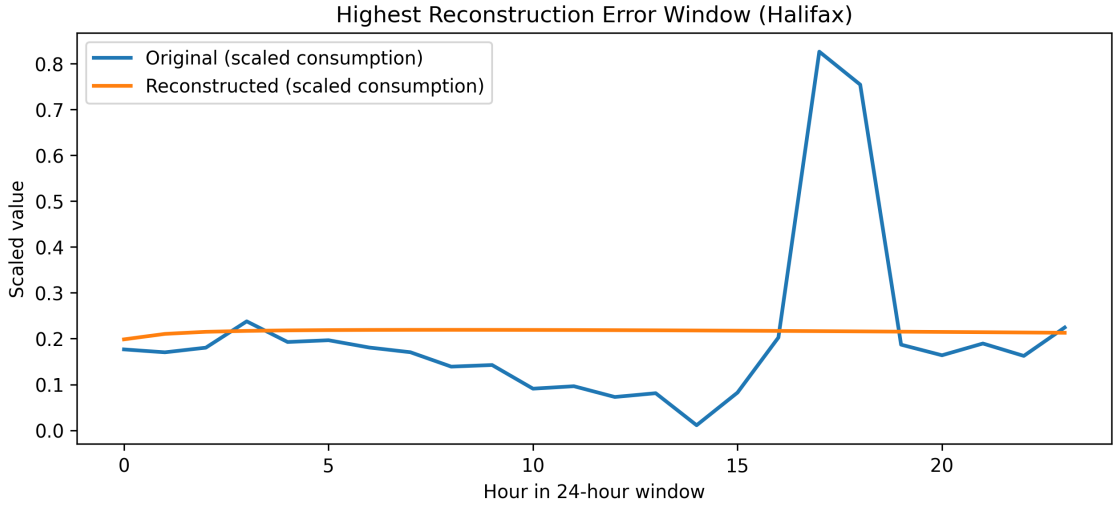

Figure 4: Autoencoder reconstruction example for highest-error sequence window.

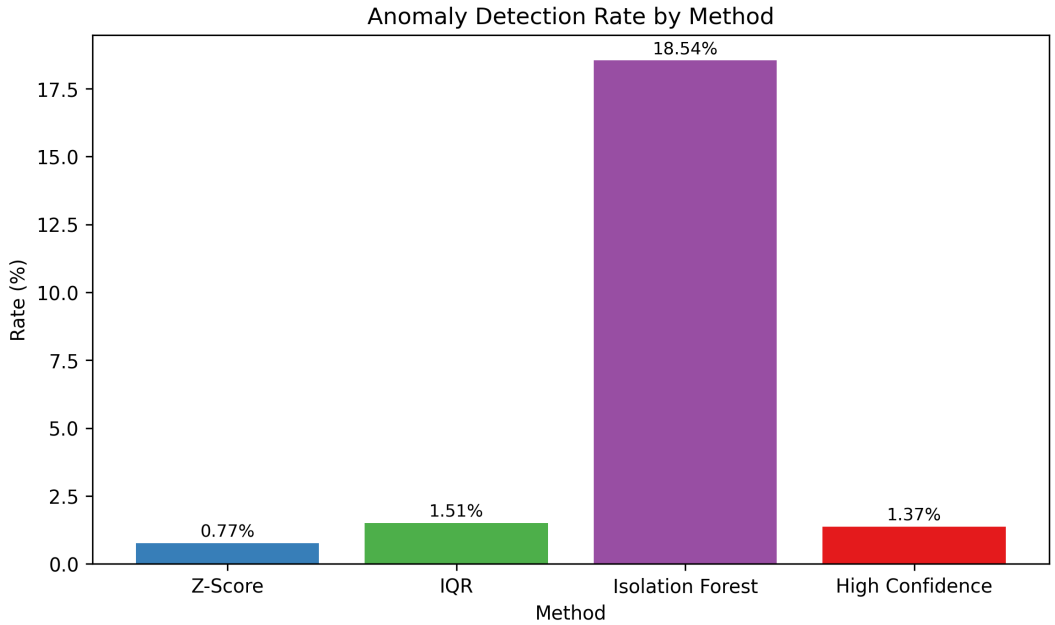

Figure 5: Intermediate anomaly detection rate comparison.

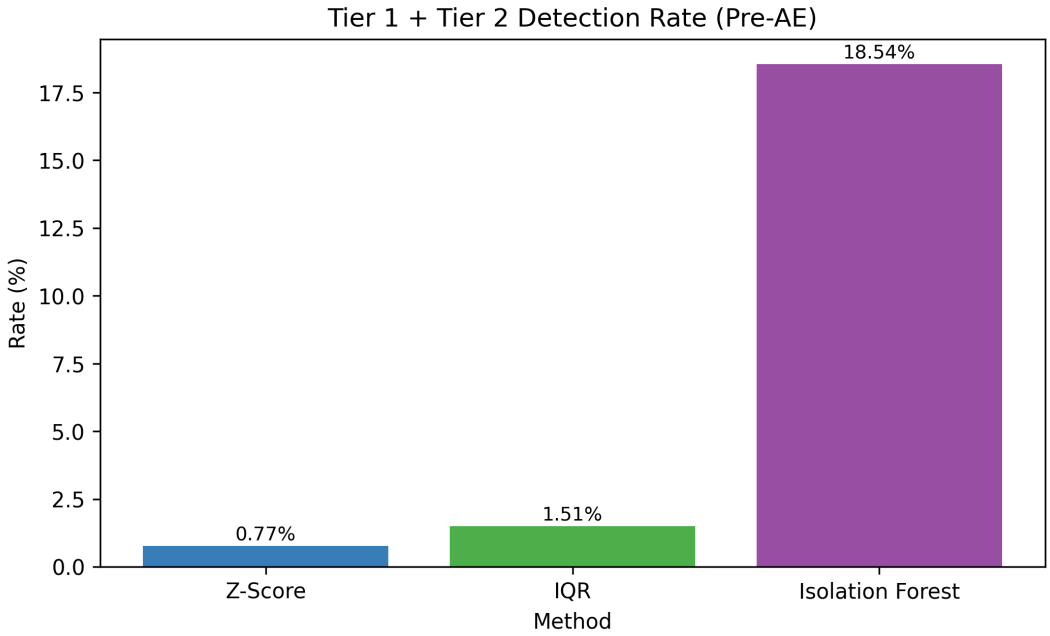

Figure 6: Tier 1 and Tier 2 detector comparison (pre-autoencoder).

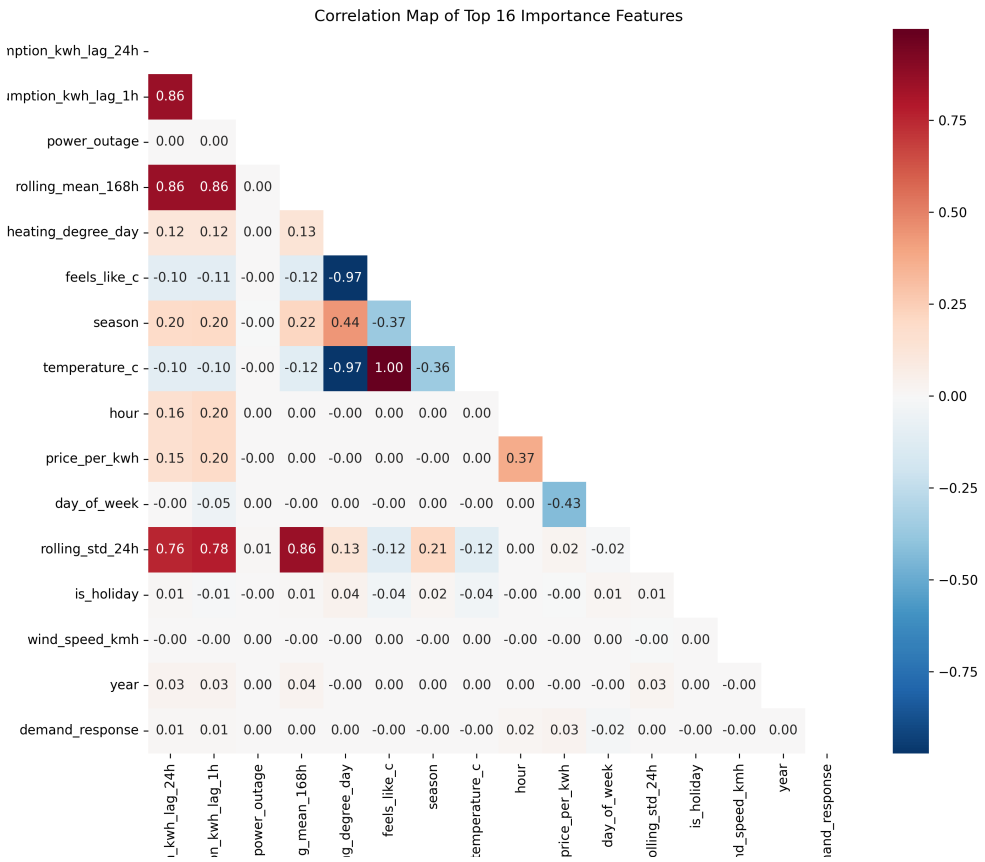

Figure 7: Feature correlation heatmap from engineered feature space.

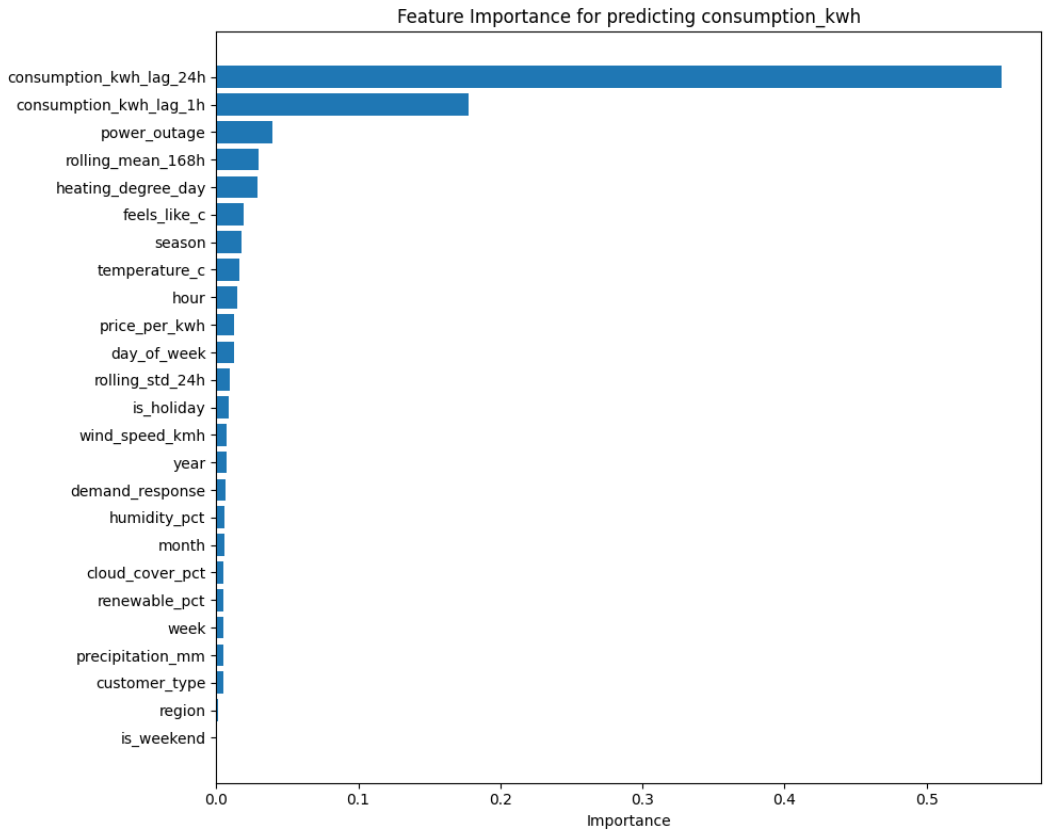

Figure 8: Feature importance output (configuration 25).

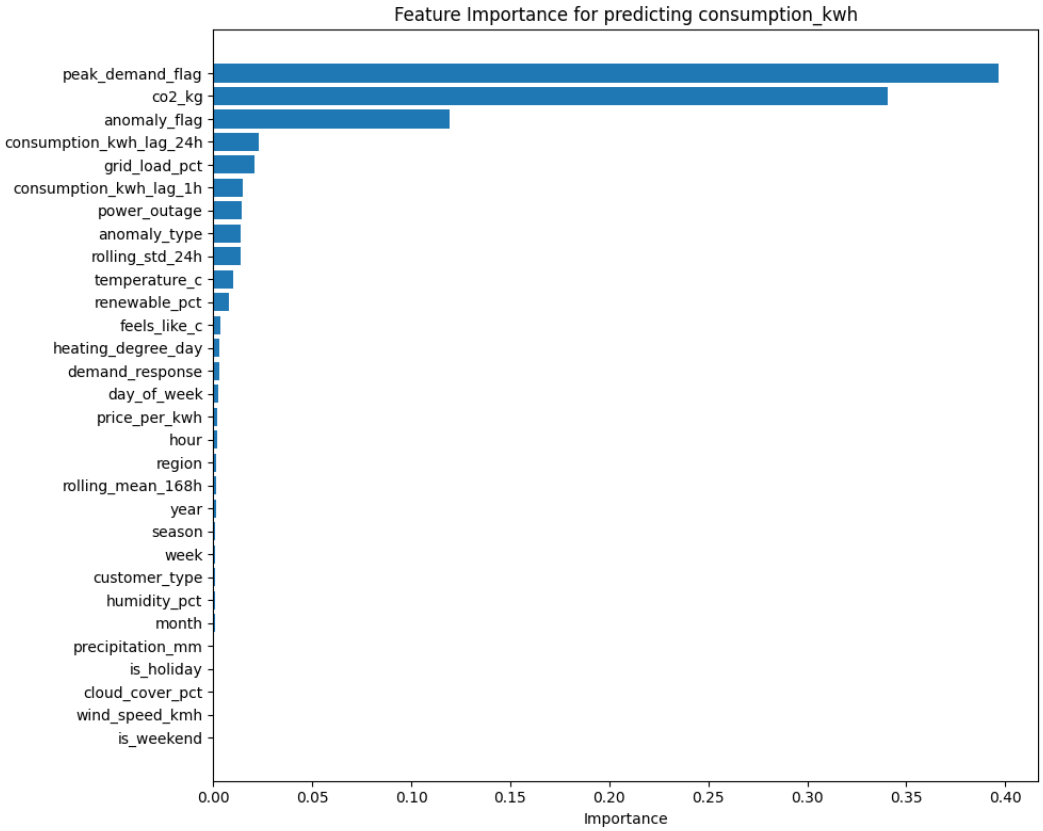

Figure 9: Feature importance output (configuration 30).

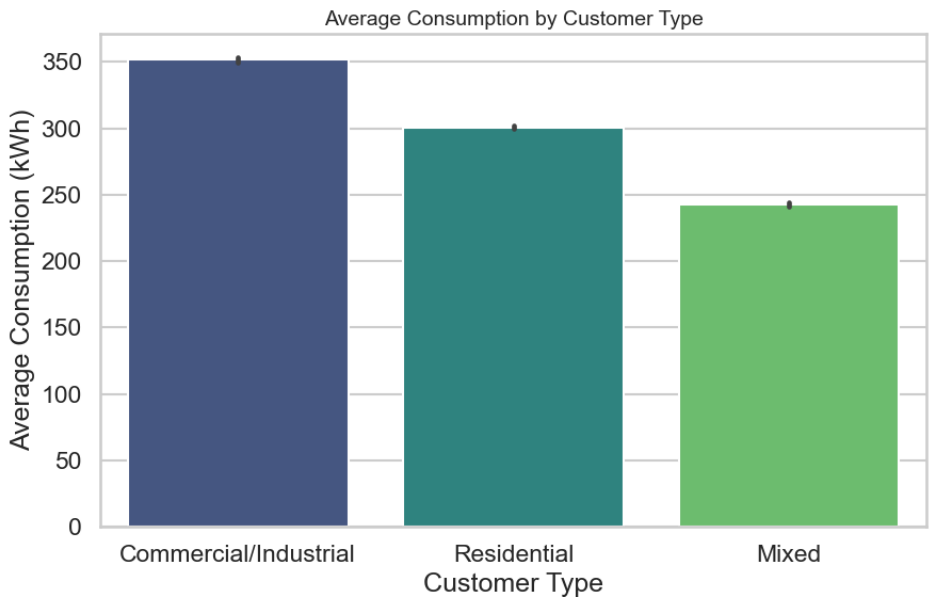

Figure 10: Customer-type versus consumption relationship.

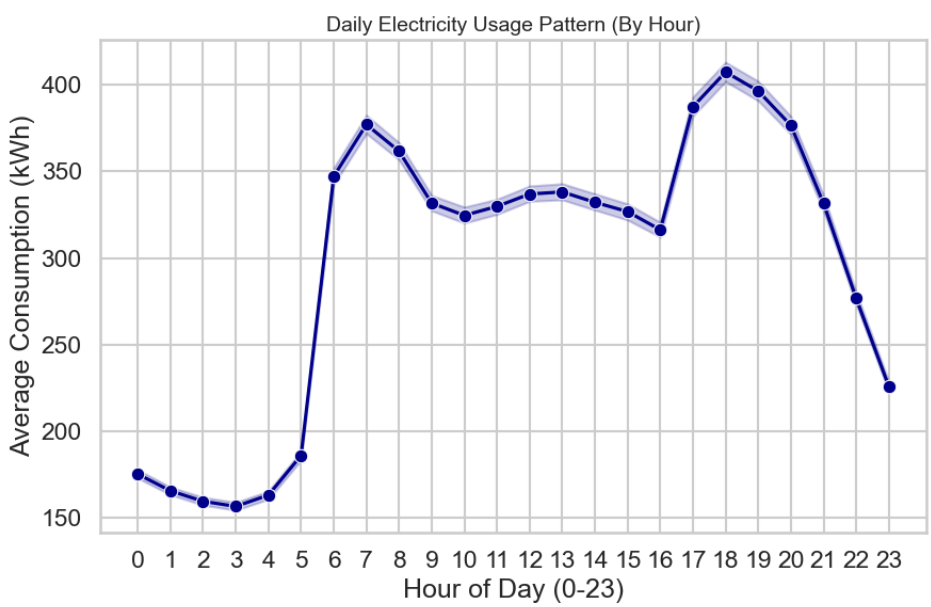

Figure 11: Hourly consumption pattern visualization.

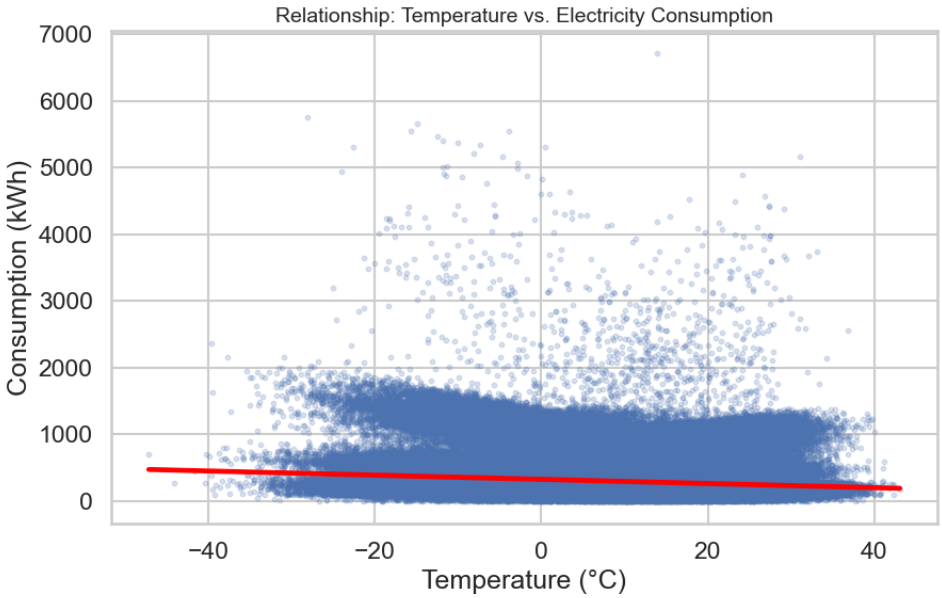

Figure 12: Temperature versus consumption relationship.

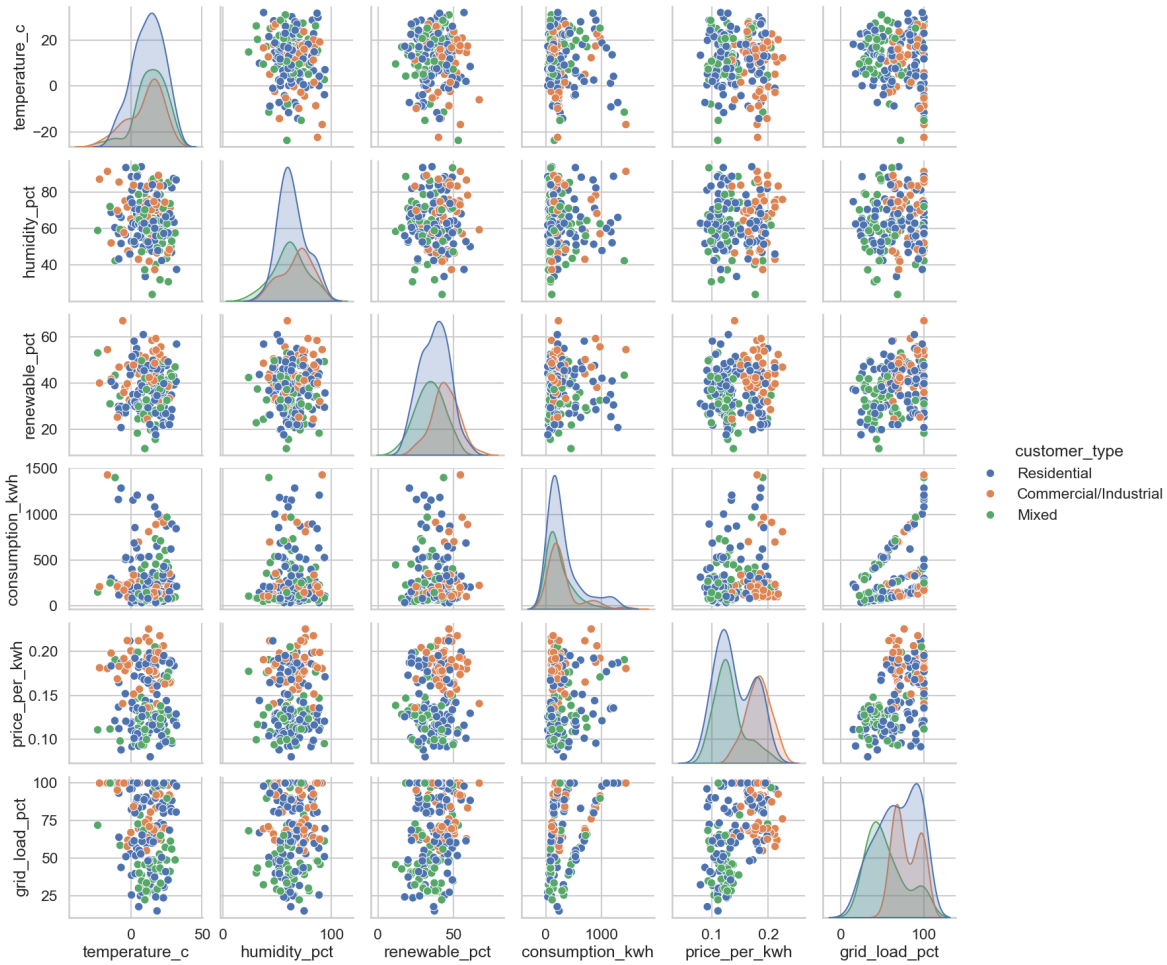

Figure 13: Multivariate pair plot used for exploratory assessment.

Bibliography